Some model tests

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Abstract

Some numerical test about training and testing of cross section fitting using neural networks

I. INTRODUCTION

We use a neural network as universal interpolator to fit calculations of total cross sections (TCS) within two theoretical models, namely the Continuum-Distorted-Wave with Eikonal-Initial -State (CDW-EIS) and the Classical Trajectory MonteCarlo (CTMC) for ionization of hydrogen by bare ion impact. Here we consider bare projectiles with charges Z_P ranging between 1 and 9

II. METHODS

A. The training set

The training set consist in a number of 1013 TCS. We have calculated 513 TCS with the CDW-EIS model and the CTMC model as it is specified in Table I. It is important to note that we have considered here the semiempiric model of Rudd (RSEM) [poner cita] on an equal footing with the theoretical models.

Theory	Number of Impact Energies	Impact Energy Range (keV/amu)	Ion charges
CDW-EIS	37	10 - 100000	1 - 9
CTMC	20	15 - 2500	1 - 9
Semi Empiric	500	10 - 5000	1

TABLE I. Number and energy range of TCS calculated for each model.

The input variables are, the impact energy, the projectile charge, and the theory used to obtain the TCS. Since MLPs are not suitable to handle very tiny numbers nor several orders of magnitude, as in the case of this training set, we must perform a preprocessing stage that include the data compression by taking the decimal logarithm of the impact energy and the TCS. These preprocessing allow us to avoid gradient vanishing in the backpropagation algorithm used to fit the data because using tiny floating point values for the energy and the TCS. Moreover, logarithm produce a compactacion of the several magnitude orders

that TCS present for the wide impact energies considered here. On the other hand, the categorical variable corresponding to the theory must be casted to a numerical value. We choose to expand a three dimensional One Hot Encoding for the three theories considered here.

B. The model

We show in Figure 1 a scheme of the model used in this work. After the preprocessing stage we have a MLP with a 5-dimensional input, then the hidden layers (N_L) and a single output corresponding to the $\log_{10}$ of the TCS.

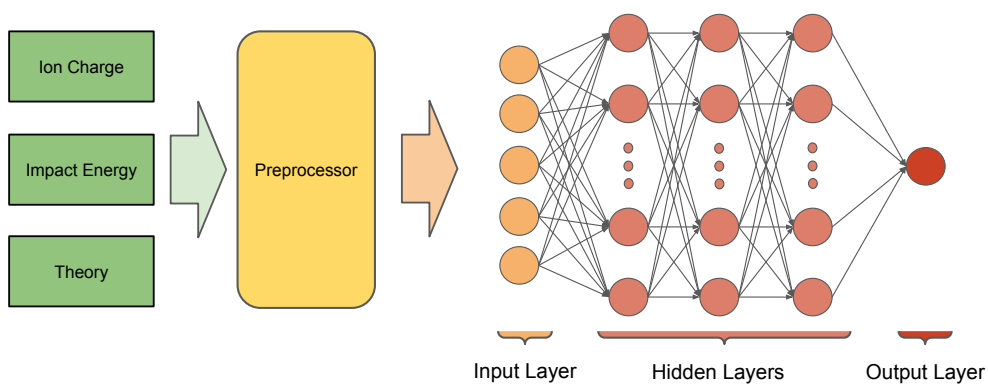

FIG. 1. Interpolator scheme

To find the best architecture of the MLP, as the training set is not big enough, we have performed averaging over a k-fold split over the training set, and the corresponding training for models with hidden layers ranging from one to five ($N_L = 1, ...5$), with different number of units per layer ($N_H = 2^n$ for $n = \{4, ..., 8\}$). As it is well known, to introduce non linearity in the MLPs, non linear activation functions must be used for each unit output. Unlike the usual trend to use the fast-to-compute Rectifier Linear Unit activation function (ReLU), in this work we use two kind of infinite differentiable functions: the hyperbolic tangent, and a gaussian activation. The choice of smooth activation functions can be easily justified by the fact that TCS are smooth functions of the impact energy, and extreme non differentiable oscillating fits caused by ReLU and its variations are naturally avoided.

This fitting process is a typical supervised learning. In supervised learning some *distance*

function between the model prediction y_{pred} and the true value y_{true} must be computed. This *distance* function, is called the loss function $\mathcal{L}_\theta$, where θ refer to the MLP parameters. In this work we explore two loss functions

$$\mathcal{L}_\theta^{LCH}(y_{true}, y_{pred}) = \log[\cosh(y_{pred} - y_{true})] \tag{1}$$

$$\mathcal{L}_\theta^{ERR}(y_{true}, y_{pred}) = L_1(y_{pred}, y_{true}) + L_2(y_{pred}, y_{true}) + Rel(y_{pred}, y_{true}) \tag{2}$$

In Eq. (2) three different metrics are added,

$$L_1(y_{true}, y_{pred}) = \frac{1}{N}\sum_{i=1}^{N} |y_{pred} - y_{true}| \tag{3}$$

$$L_2(y_{true}, y_{pred}) = \frac{1}{N}\sum_{i=1}^{N} (y_{pred} - y_{true})^2 \tag{4}$$

$$R(y_{true}, y_{pred}) = \frac{1}{N}\sum_{i=1}^{N} \left|\frac{y_{pred} - y_{true}}{y_{true}}\right|. \tag{5}$$

Usually, L_1, L_2, and R are named as the mean absolute error (MAE), the mean squared error (MSE) and the mean relative error (MRE), respectively. We combine these three measures in order to guide the optimization algorithm for the different ranges in the $\log_{10}(TCS)$. While on one hand MSE ponderates big differences, especially where target values are higher, the MAE does it at intermediate regions. On the other hand, R runs over whole range on equal footing.

As we are dealing with analytic activation functions, we can use the gradient descent algorithm or any variant of it. In this work we use the Adam algorithm with an initial learning rate of $lr = 1 \times 10^{-4}$ to *backpropagate* the model and update the parameters.

First, we want to point out that the increase of N_L above two hidden layers does not produce significant improvements. A similar behavior occurs with the increasing of N_H above $N_H = 128$, as can be observed in Fig. 2. In this Figure we show the evolution of both loss function during the hyperparamenter exploration process as function of the iteration number or *epoch* of the optimization process. Moreover, we observe that $\mathcal{L}_\theta^{ERR}$ produces a more soft optimization curves, and, except for the case of $N_H = 32$ with tanh activation function in panel (d) this loss does not produce overfitting or an increase of the loss during the optimization process. The opposite behavior can be observed for $\mathcal{L}_\theta^{LCH}$ in panels (c), (e) and (g). In the following, we use $\mathcal{L}_\theta^{ERR}$ as the loss function. We also observe that gaussian

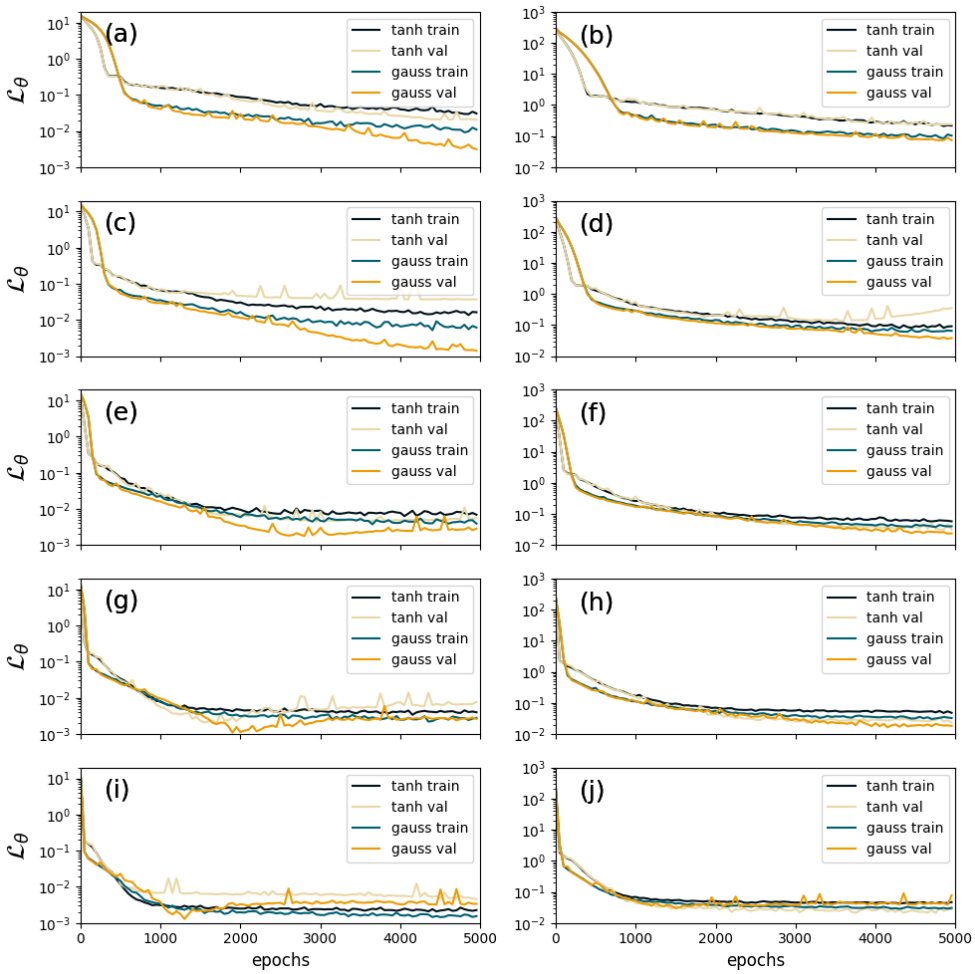

FIG. 2. Averaged in 5-folded loss functions calculated over the training and the validation sets as function of the epoch. In all the cases the number of layesrs $N_L = 2$. Left column: $\mathcal{L}_\theta^{LCH}$ loss function, right column: $\mathcal{L}_\theta^{ERR}$ loss function. Each row correspond to architectures with increasing number of hidden units $N_H = \{16, 32, 64, 128, 256, 512\}$, shown in panels (a) and (b); (c) and (d); (e) and (f); (g) and (h); and (i) and (j), respectively.

activation is more adequate. To examinate the behavior of the three terms of $\mathcal{L}_\theta^{ERR}$ we show in Fig. 3 the evolution during the training process for mean error functions over the validation set. The error computed are MAE (panels (a) and (b)), RMSE in panels (c) and (d), and MRE shown in panels (e) and (f). The left column correspond to tan activation, meanwhile the right column to gaussian activation function respectively finding a behavior that confirms the use of $N_H = 128$.

To get a deep understand of the activation function we explore the evolution of the

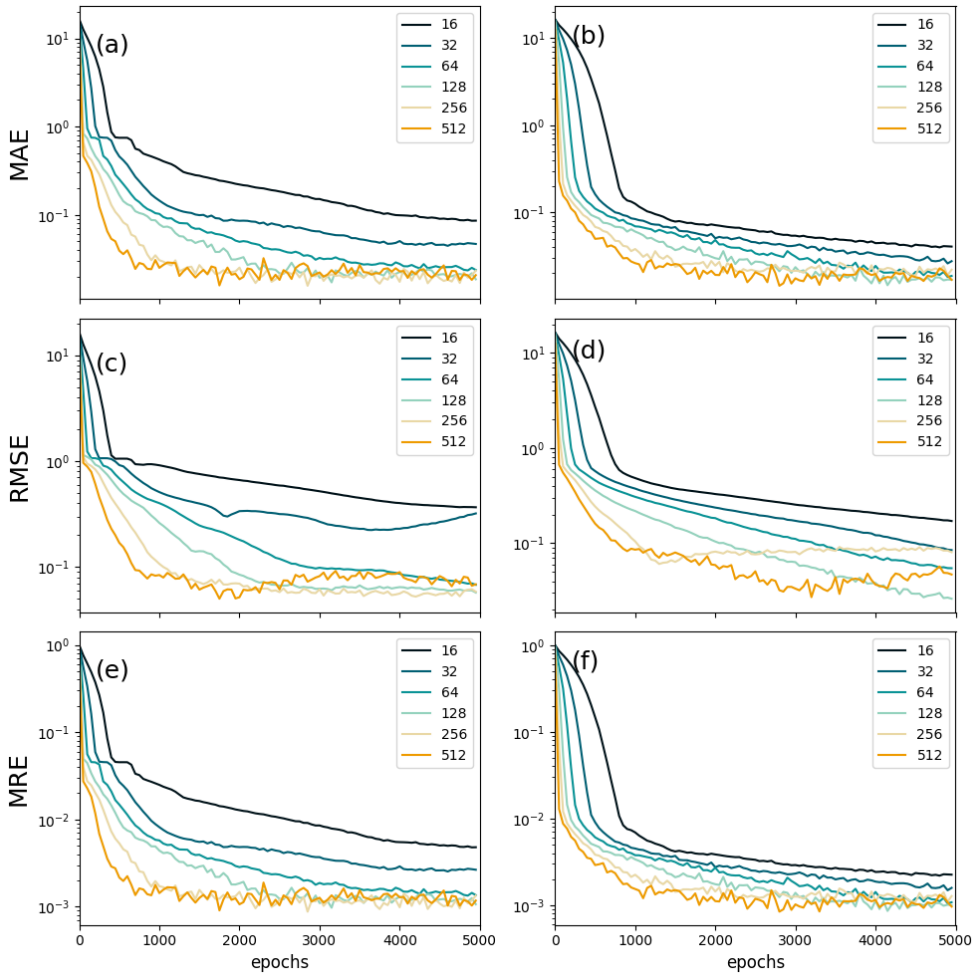

FIG. 3. Metrics over validation set for different network architectures. Left column (a, c, and e) metrics with tanh activation function. Right column (b-d-f) gaussian activation function. Each row correspond to a different metric over the $\log_{10}$ of the TCS.

maximum of the errors instead of mean values in Fig. 4. In this Figure we observe that gaussian activation exhibits monotonically decreasing errors for $N_H \leq 128$ hidden units, in contrast to tanh activation which shows more erratic behavior.

With the above results we decide to use as the optimum model a MLP with two hidden layers with 128 units each, using a gaussian activation function. This model has a total of 17409 trainable parameters.

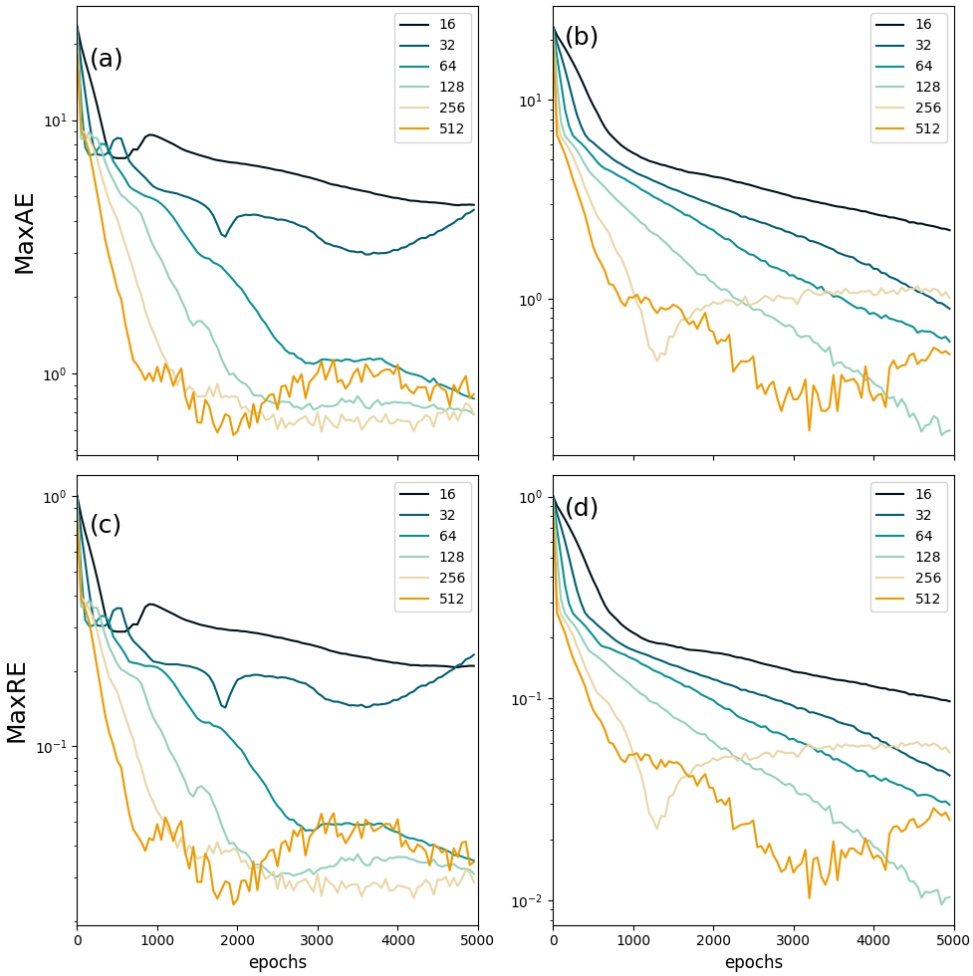

FIG. 4. Maximum values of metrics over validation set for different network architectures. Left column (a, c) metrics with tanh activation function. Right column (b-d) gaussian activation function. Each row correspond to a different metric over the $\log_{10}$ of the TCS.

III. RESULTS

In this work we fit total cross sections calculated by means of the CDW-EIS, CTMC, and the RSEM for ionization of hydrogen by bare projectiles with charges ranging from 1 to 9.

This section can be divided in two main parts: one with the fits for the TCS for each case and the other with numerical experiments retraining models with missing data for particular projectile charges.

A. Evaluation of the fits

We perform evaluations over the whole TCS training set with the best model found in the previous section to understand the different errors in cm^2. It is important to note that we have interpolated the $\log_{10}(TCS)$, which lead to errors different to those shown in the previous section.

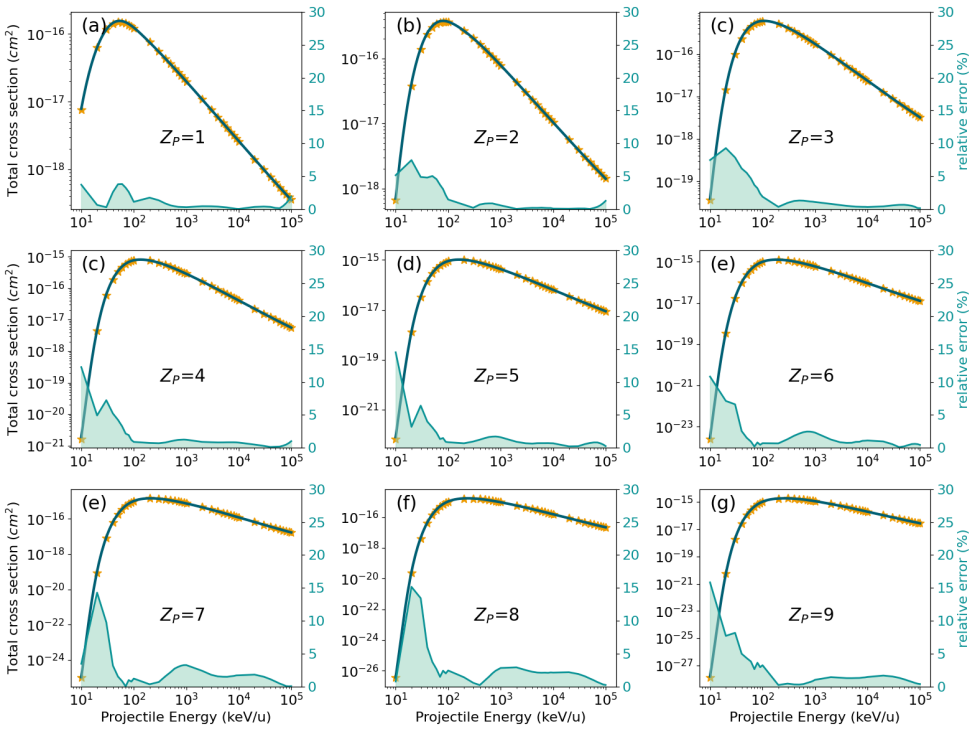

FIG. 5. Solid lines: Prediction of the MLP for CDW-EIS calculations as a function of the projectile impact energy for different Z_p. Orange stars: calculated TCS. Panels (a) to (g) correspond to $Z_P = 1$ to $Z_P = 9$ respectively. Relative error is drawn as green filled curve. The right axis scale correspond to the percentage relative error calculated at the ground truth values of the TCS.

We show in Figures 5 and 6 the performance for MLP to fit the calculated data. We observe relative errors below five percent for high energies where the cross sections present smooth changes. In the low energy region where TCS changes several orders of magnitude the error raises up to 15%

- This plot shows the power of gaussian activation.

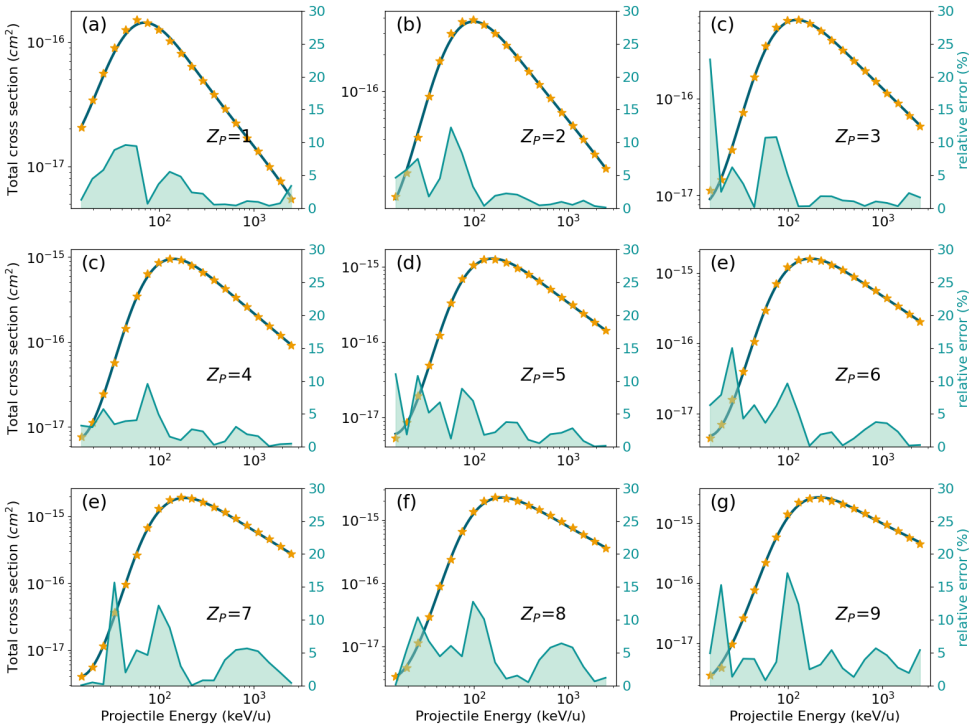

FIG. 6. Prediction of the MLP and percentage relative errors as in Fig. 5 for CTMC calculations as a function of the projectile impact energy for different Z_p.

IV. DISCUSSION

- WE NEED TO DO THE EXPERIMENT WITH ZP HOLES AND ENERGY EXTRAPOLATIONS

- WE NEED TO DEFINE RANGES OF INPUT ENERGIES IN THE INPUT PIPELINE

- SEBASTIAN HAS TO PLOT THE SEMIEMPIRIC FIT!!!!

- MISSING THE BENCHMARKING AGAINST THE CUBIC INTERPOLATOR. ASK TO MICHELE IN DEEP.

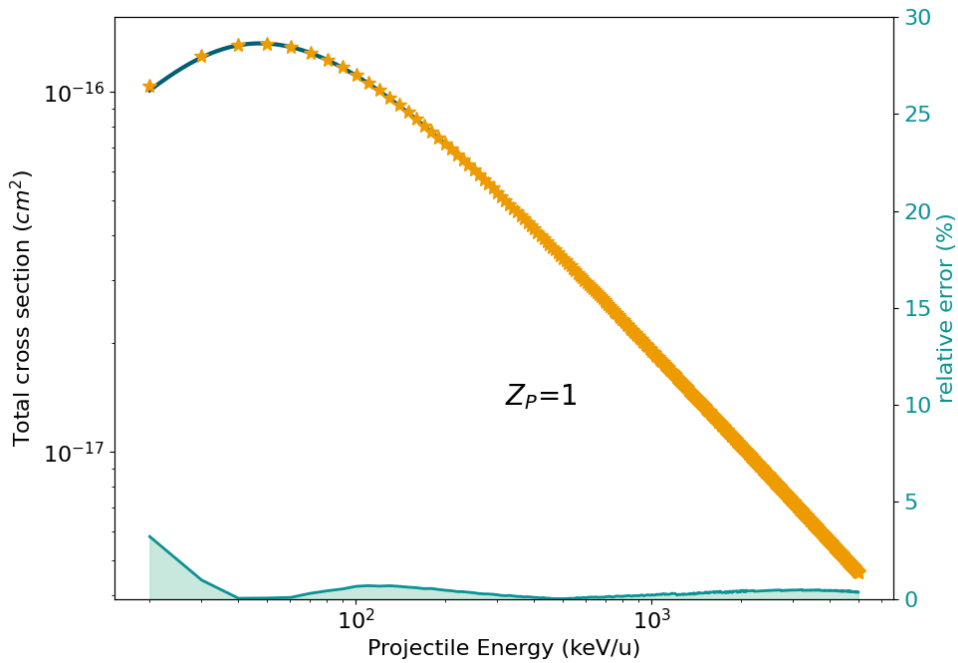

FIG. 7. Prediction of the MLP and percentage relative errors as in Fig. 5 for RSEM calculations as a function of the projectile impact energy for $Z_p = 1$.

V. CONCLUSION

Summarize your conclusions and future work.

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[1] Author, "Title," *Journal*, vol. X, pp. XX-XX, Year.